THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of all two-digit multiples of 6 is: 1

(a) 816 **(b)** 714 **(c)** 798 **(d)** 810

2. The sum of the first 16 terms of the AP $9, 16, 23, 30, \ldots$ is: 1

(a) 1105 **(b)** 984 **(c)** 870 **(d)** 1968

3. The middle term of an AP with 15 terms, first term 10 and last term 80, is: 1

(a) 45 **(b)** 50 **(c)** 40 **(d)** 80

4. If $a_9 = 33$ and $d = 4$, then $a =$ 1

(a) 65 **(b)** -3 **(c)** 1 **(d)** 5

5. The 14th term of the AP whose 1st term is 9 and 5th term is 25 is: 1

(a) 57 **(b)** 61 **(c)** 56 **(d)** 65

6. The number of terms in the AP $7, 16, \ldots, 169$ is: 1

(a) 18 **(b)** 21 **(c)** 20 **(d)** 19

7. The sum of all two-digit multiples of 7 is: 1

(a) 728 **(b)** 714 **(c)** 630 **(d)** 735

8. The sum of the first 11 terms of the AP $2, 11, 20, 29, \ldots$ is: 1

(a) 618 **(b)** 1034 **(c)** 425 **(d)** 517

9. The middle term of an AP with 17 terms, first term 3 and last term 99, is: 1

(a) 51 **(b)** 57 **(c)** 45 **(d)** 99

10. If $a_7 = 31$ and $d = 3$, then $a =$ 1

(a) 49 **(b)** 16 **(c)** 10 **(d)** 13

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum of all natural numbers between 5 and 150 that are divisible by 6. 2

12. Which term of the AP $30, 28, 26, 24, \ldots$ is 8? 2

13. Find the sum of the first 15 terms of the AP $27, 25, 23, 21, \ldots$ 2

14. Find the sum of all natural numbers between 20 and 300 that are divisible by 7. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an AP, the 5th term is 38 and the 10th term is 83. Find the AP and its 13th term. **3**

16. Find the sum of the first 10 terms of the AP whose nth term is $a_n = 6n - 4$. **3**

17. The sum of the first 8 terms of an AP is 172 and the first term is 4. Find d and the 8th term. **3**

OR

In an AP, the 2nd term is 19 and the 11th term is 55. Find the AP and its 18th term. **3**

18. Find the sum of the first 16 terms of the AP whose nth term is $a_n = 7n + 3$. **3**

OR

The sum of the first 16 terms of an AP is 472 and the first term is 7. Find d and the 16th term. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. How many terms of the AP $6, 11, 16, 21, \ldots$ must be taken so that the sum is 696? **5**

OR

A ladder has 17 rungs whose lengths decrease uniformly from 75 cm at the bottom to 35 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. **5**

20. How many terms of the AP $60, 55, 50, 45, \ldots$ must be taken so that the sum is 360? **5**

OR

A ladder has 19 rungs whose lengths decrease uniformly from 66 cm at the bottom to 30 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. **5**

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND